

SYLLABUS

Bit Manipulation: Introduction, Applications: Counting Bits, Palindrome Permutation, Remove All Ones with Row and Column Flips, Encode Number, Swapping Numbers, Find the Non Repeated Elements in Array, Minimum Bit Flips to Convert Number, Longest Nice Subarray.

Introduction

In programming, an n-bit integer is internally stored as a binary number that consists of n bits. For example, the Java type int is a 32-bit type, which means that every int number consists of 32 bits. For example, the bit representation of the int number 43 is

000000000000000000000000101011.

The bits in the representation are indexed from right to left.

To convert a bit representation $b_k \dots b_2b_1b_0$ into a number, the formula

$b_k2^k + \dots + b_22^2 + b_12^1 + b_02^0$ can be used.

For example,

$$1 \cdot 2^5 + 1 \cdot 2^3 + 1 \cdot 2^1 + 1 \cdot 2^0 = 43.$$

- The bit representation of a number is either signed or unsigned.
- Usually a signed representation is used, which means that both negative and positive numbers can be represented.
- A signed variable of n bits can contain any integer between 2^{n-1} and $2^{n-1} - 1$. For example, the int type in Java is a signed type, so an int variable can contain any integer between -2^{31} and $2^{31} - 1$.
- The first bit in a signed representation is the sign of the number (0 for non negative numbers and 1 for negative numbers), and the remaining $n - 1$ bits contain the magnitude of the number.
- Two's complement is used, which means that the opposite number of a number is calculated by first inverting all the bits in the number and then increasing the number by one. For example, the bit representation of the number -43 is 1111111111111111111111111111010101.
- There is a connection between the representations: a signed number $-x$ equals an unsigned number $2^n - x$. For example, the signed number $x = -43$ equals the unsigned number $y = 2^{32} - 43$.
- If a number is larger than the upper bound of the bit representation, the number will overflow.
- In a signed representation, the next number after $2^{n-1} - 1$ is -2^{n-1} .
- Initially, the value of x is $2^{31} - 1$. This is the largest value that can be stored in an int

For example, $14 \prec 2 = 56$, because 14 and 56 correspond to 1110 and 111000.

Similarly, $49 \gg 3$ is 6, because 49 and 6 correspond to 110001 and 110.

$x \ll k$ corresponds to multiplying x by 2^k , and

$x \gg k$ corresponds to dividing x by 2^k rounded down to an integer.

What is Bit Manipulation?

Bit manipulation is the process of applying logical operations on a sequence of bits, the smallest form of data in a computer, to achieve a required result. Bit manipulation has constant time complexity and process in parallel, meaning it is very efficient on all systems.

Most programming languages will have you work with abstractions, like *objects* or *variables*, rather than the bits they represent. However, direct bit manipulation is needed to improve performance and reduce error in certain situations.

Bit manipulation requires a strong knowledge of *binary* and *binary conversion*.

Here's a few examples of tasks that require bit manipulation:

- Low-level device control
- Error detection and correction algorithms
- Data compression
- Encryption algorithms
- Optimization

For example, take a look at the difference between an arithmetic and bit manipulation approach for finding the green portion of an RGB value:

```
//arithmetic  
(rgb / 256) % 256
```

```
// bit  
(rgb >> 8) & 0xFF
```

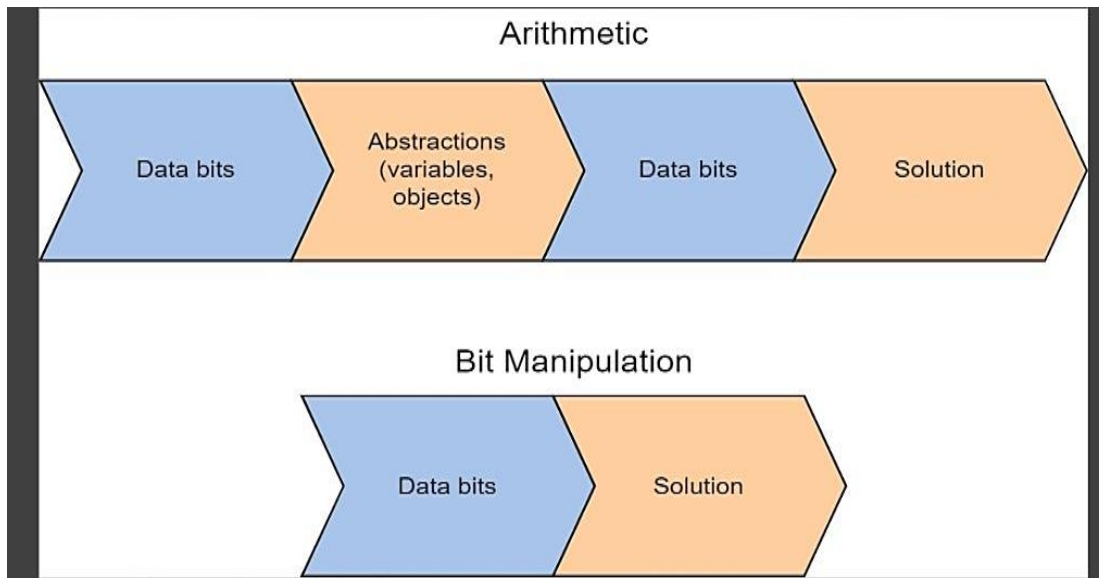
While both do the same thing, the second option is considerably faster, as it works directly within memory rather than through a level of abstraction.

We'll explore what each of these operators do later in this article ($\gg$ and $\&$).

Bitwise Operators

Bitwise operations take one or more bit-patterns or binary numerals and **manipulate them at the bit level**. They're essentially our tool to manipulate bits to achieve our operations.

While arithmetic operations perform operations on human-readable values (1+2), bitwise operators manipulate the low-level data directly.



Difference in steps between arithmetic and bit operations

Advantages

- They are fast and simple actions.
- They are directly supported by the processor.
- They are used to manipulate values for comparisons and calculations.
- Bitwise operations are incredibly simple and faster than arithmetic operations.

List of Bitwise operators

Operator	Name of Operator	Usage
&	Bitwise AND	Used to mask particular part of byte
	Bitwise OR	
~	One's complement/NOT	Used to turn a bit on/off
^	Bitwise XOR	
<<	Left Shift	Used to shift the bit to the left
>>	Right Shift	Used to shift the bit to the right

Bitwise AND [&]:

AND (&) is a binary operator that compares two operands of equal length. The operands are converted from their readable form to binary representation. For each bit, the operation checks if both bits are 1 across both operands. If yes, that bit is set to 1 in the answer. Otherwise, the corresponding result bit is set to 0.

It essentially multiplies each bit by the corresponding bit in the other operand. As multiplying anything by 0 results in 0, the AND comparison with any 0 bit will result in 0.

If two input bits are 1, the output is 1.

In all other cases its 0, for example:

1 & 0 => yields to 0.

0 & 1 => yields to 0.

0 & 0 => yields to 0.

0101 (decimal 5) AND 0011 (decimal 3)

= 0001 (decimal 1)

The operation may be used to determine whether a particular bit is set (1) or clear (0).

It's also used to clear selected bits of a register in which each bit represents an individual Boolean state.

Example:

```
class AndOperation
{
    public static void main( String args[] )
    {
        int x = 12;
        int y = 10;
        System.out.println("Bitwise AND of (" + x + " , " + y + ") is: " + (x & y)); // yields to 8
    }
}
```

Output: Bitwise AND of (12 , 10) is: 8

Bitwise OR(|):

The OR operator (|) is a binary operator that takes two equal-length operands but **compares them** in the opposite way to AND; if either corresponding bit is 1, the answer is 1. Otherwise, the answer will be 0. In other words, Bitwise OR returns '1' if one of the inputs given is 1.

If two input bits are 0, the output is 0.

In all other cases, it is 1. For example:

1 | 0 => yields to 1.

0 | 1 => yields to 1.

1 | 1 => yields to 1.

a = 12

b = 10

a in Binary : 0000 0000 0000 1100

b in Binary : 0000 0000 0000 1010

a | b : 0000 0000 0000 1110

This is often used as an interim logic step for solving other problems.

Example:

```
class OROperation
{
    public static void main(String[] args)
    {
        int x = 12;
        int y = 10;
        System.out.println("Bitwise OR of " + x + ", " + y + " is: " + (x | y));
    }
}
```

Output:

Bitwise OR of 12, 10 is: 14

NOT (~) Operator

NOT (~), or sometimes called the bitwise complement operator, is a unary operation that takes a single input and **swaps each bit** in its binary representation to the opposite value.

All instances of 0 become 1, and all instances of 1 become 0. In other words, NOT inverts each input bit. This inverted sequence is called the **one's complement** of a bit series.

For example, consider x = 1

The binary number representation of x is:

x=00000000 00000000 00000000 00000001

Now, Bitwise NOT of x will be:

x=11111111 11111111 11111111 11111110

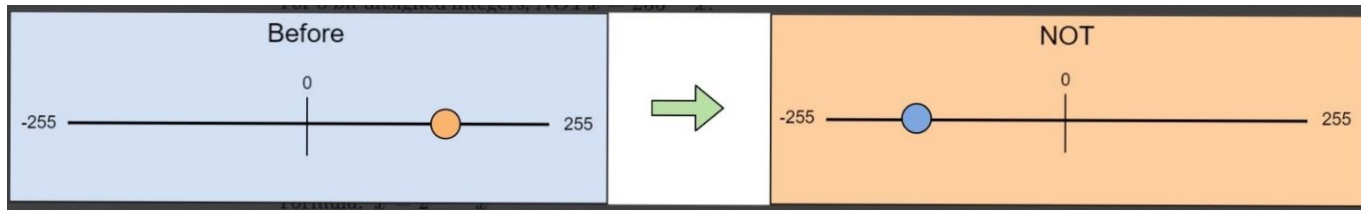
So: x contains 31 zeros(0's) and one 1

~x contains 31 ones(1's) and one 0(zero)

This makes the number negative as any bit collection that starts with 1 is negative.

NOT is useful for flipping unsigned numbers to the mirrored value on the opposite side of their mid-point.

For 8-bit unsigned integers, $NOTx = 255 - x$.



Formula: $x = 2^{32} - x$

Example:

```
class NOTOperation
{
    public static void main( String args[] )
    {
        int a = 1;
        System.out.println("Bitwise NOT of a is : " + ~a);
    }
}
```

Output:

Bitwise NOT of a is : -2

Explanation:

a = 1 in binary is

00000000 00000000 00000000 00000001

~a = 11111111 11111111 11111111 11111110

Since leftmost bit is 1, which means -ve number, to get real number we invert 1 to 0 and 0 to 1 and add 1

= 00000000 00000000 00000000 00000001 + 1 = 2

Bitwise Exclusive OR (XOR) (^) Operator

The bitwise XOR operation (^), short for “Exclusive-Or”, is a binary operator that takes two input arguments and **compares each corresponding bit**. If the bits are opposite, the result has a 1 in that bit position. If they match, a 0 is returned.

$1 \wedge 1 \Rightarrow$ yields to 0.

$0 \wedge 0 \Rightarrow$ yields to 0.

$1 \wedge 0 \Rightarrow$ yields to 1.

$0 \wedge 1 \Rightarrow$ yields to 1.

For example:

a = 12

b = 10

a in binary : 0000 0000 0000 1100

b in binary : 0000 0000 0000 1010

a ^ b : 0000 0000 0000 0110

XOR is used to invert selected individual bits in a register or manipulate bit patterns that represent Boolean states.

XOR is also sometimes used to set the value of a registry to zero as XOR with two of the same input will always result in 0.

Example:

```
class XOROperation
{
    public static void main(String args[])
    {
        int x = 12;
        int y = 10;
        System.out.println("Bitwise XOR of (x , y) is : " + (x ^ y));
    }
}
```

Output:

Bitwise XOR of (x , y) is : 6

Bitwise Operators comparison

Below is a table showing a comparison of results of all the bitwise operators mentioned above based on different values of the compared bits (A and B).

A	B	A & B	A B	A ^ B
1	0	0	1	1
0	1	0	1	1
1	1	1	1	0
0	0	0	0	0

Bitwise Shift Operators

Left and right shift operator

A bit shift is a Bitwise operation where the order of a series of bits is moved to efficiently perform a mathematical operation. A bit shift moves each digit in a number's binary representation left or right by a number of spaces specified by the second operand.

These operators can be applied to integral types such as int, long, short, byte, or char.

There are three types of shifts:

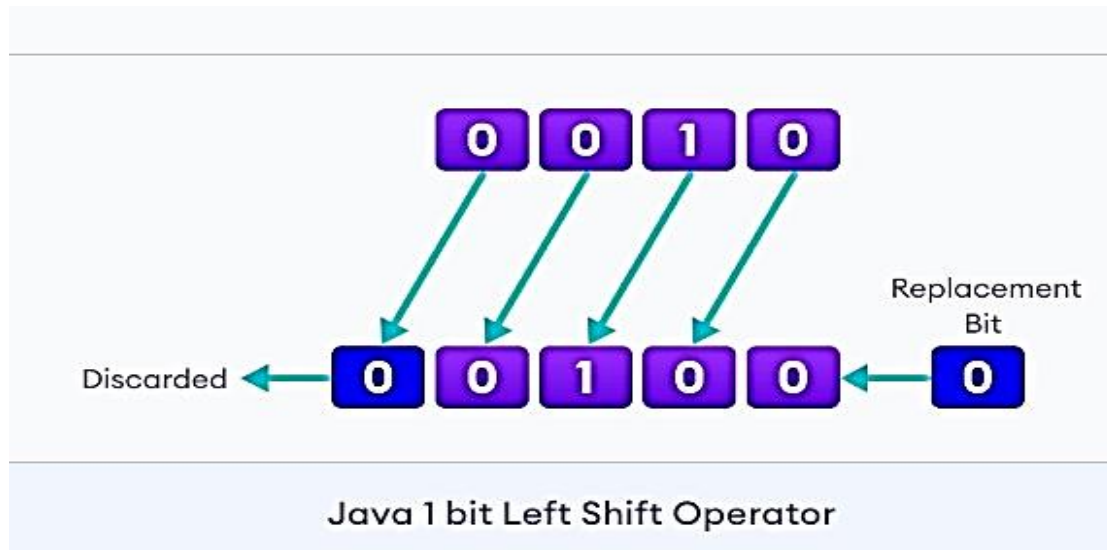
Left shift: << is the left shift operator and meets both logical and arithmetic shifts' needs.

Arithmetic/signed right shift: >> is the arithmetic (or signed) right shift operator.

Logical/unsigned right shift: >>> is the logical (or unsigned) right shift operator.

In Java, all integer data types are signed and << and >> are solely arithmetic shifts.

Here's an example of a left shift:



6=00000000 00000000 00000000 00000110

Shifting this bit pattern to the left one position ($6 \ll 1$) results in the number 12:

$6 \ll 1 = 00000000 00000000 00000000 00001100$

As you can see, the digits have shifted to the left by one position, and the last digit on the right is filled with a zero. Note that shifting left is equivalent to multiplication by powers of 2.

$6 \ll 1 \rightarrow 6 * 2^1 \rightarrow 6 * 2$

$6 \ll 3 \rightarrow 6 * 2^3 \rightarrow 6 * 8$

Well-optimized compilers will use this rule to replace multiplication with shifts whenever possible, as shifts are faster.

Example:

```
class LeftShift
{
    private static int helper(int number, int i)
    {
        return number << i;    // multiplies `number` with 2^i times.
    }
    public static void main(String[] args)
    {
        int number = 100;
        System.out.println(number + " shifted 1 position left, yields " + helper(number, 1));
        System.out.println(number + " shifted 2 positions left, yields " + helper(number, 2));
        System.out.println(number + " shifted 3 positions left, yields " + helper(number, 3));
        System.out.println(number + " shifted 4 positions left, yields " + helper(number, 4));
    }
}
```

Output:

100 shifted 1 position left, yields to 200

100 shifted 2 positions left, yields to 400
100 shifted 3 positions left, yields to 800
100 shifted 4 positions left, yields to 1600

With right shift, you can either do arithmetic ($\gg$) or logical ($\gg$) shift.

The difference is that arithmetic shifts maintain the same most significant bit (MSB) or **sign bit**, the leftmost bit which determines if a number is positive or negative.

$110110101 \gg 1 = 1101\ 1010$.

Formula: $x \gg y = x/2^y$

On the other hand, a logical shift simply moves everything to the right and replaces the MSB with a 0.

$10110101 \ggg 1 = 01011010$

Formula: $a \ggg b = a/2^b$

Example:

```
class RightShift
{
    public static void main(String[] args)
    {
        int number1 = 8;
        int number2 = -8;
        // 2-bit signed right shift
        System.out.println(number1 >> 2); // prints 2
        System.out.println(number2 >> 2); // prints -2
    }
}
```

Output:

2
-2

Examples on Bit manipulation:

Using Bitwise AND : *Check if a number is even*

This one tests your knowledge of how AND works and how even/odd numbers differ in binary. You can simply use:

$(x \& 1) == 0$
0110 (6)
0001 (1) &
0000 TRUE

This solution relies on two things:

2 equates to 0001

The rightmost number for all odd numbers greater than 2 is 1

Any time the final bit evaluates to 1, you know that it matched and is, therefore, an odd number. If it instead evaluates to 0, you know that no numbers matched and therefore it's even.

```
public class Test
{
    public static void main(String[] args)
    {
        int x=2;
        if((x & 1) == 0)
            System.out.println("even");
        else
            System.out.println("odd");
    }
}
```

Output:

even

Using X-OR: Java program for Convert characters to uppercase or lowercase

This trick tests your knowledge of uppercase and lowercase characters in binary. You can convert any character, ch, to the opposite case using $ch \wedge= 32$.

This is because the binary representation of lowercase and uppercase letters is nearly identical, with only 1 bit of difference.

Using the XOR operation lets us toggle that single bit and swap it to the opposite value, therefore making a lowercase character uppercase or vice versa.

Program:

```
public class Test
{
    static int x=32;
    //Toggle Case = swaps CAPS to lower case and lower case to CAPS
    static String toggleCase(char[] a)
    {
        for (int i=0; i<a.length; i++)
        {
            //Bitwise XOR with 32
            a[i]^=32;
        }
        return new String(a);
    }

    public static void main(String[] args)
    {
        String str = "CheRrY";
        System.out.print("Toggle case: ");
        str = toggleCase(str.toCharArray());
    }
}
```

```
        System.out.println(str);

        System.out.print("Original string: ");
        str = toggleCase(str.toCharArray());
        System.out.println(str);
    }
}
```

Output:

Toggle case: cHErRy
Original string: CheRrY

Using Bitwise AND: *Java program to count the number of bits set to 1 (set bits) of a given integer*

In this approach, we count only the set bits. So,

If a number has 2 set bits, then the while loop runs two times.

If a number has 4 set bits, then the while loop runs four times.

- ✓ Our while loop iterates until $n = 0$, dividing by 2 each time via the AND operator.
- ✓ On pass 1, 125 becomes 62, and count increases by 1.
- ✓ On the second pass, 62 becomes 31, and the count increases to 2.
- ✓ This continues until n becomes 0 and the count is then returned.

Program:

```
class CountSetBit
{
    private static int countSetBits(int n)
    {
        int count = 0;
        while (n > 0)
        {
            n &= (n - 1);
            count++;
        }
        return count;
    }

    public static void main(String[] args)
    {
        int number = 125;
        System.out.println("SetBit Count is : " + countSetBits(number));
    }
}
```

Output:

SetBit Count is : 6

Using Bitwise XOR: *Single Number-* Find the element in an array that is not repeated.

Input: nums = { 4, 1, 2, 9, 1, 4, 2 }

Output: 9

This solution relies on the following logic:

If we take XOR of zero and some bit, it will return that bit: $a \wedge 0 = a$

If we take XOR of two same bits, it will return 0: $a \wedge a = 0$

For n numbers, the below math can be applied: $a \wedge b \wedge a = (a \wedge a) \wedge b = 0 \wedge b = b$

For example,

$$1 \wedge 5 \wedge 1 = (1 \wedge 1) \wedge 5 = 0 \wedge 5 = 5$$

Therefore, we can XOR all bits together to find the unique number.

Program:

```
class SingleNumber
{
    private static int singleNumber(int[] nums)
    {
        int xor = 0;
        for (int num : nums)
        {
            xor ^= num;
        }
        return xor;
    }
    public static void main(String[] args)
    {
        int[] nums = {4, 1, 2, 9, 1, 4, 2};
        System.out.println("Element appearing one time is " + singleNumber(nums));
    }
}
```

Output:

Element appearing one time is 9

Using Bitwise Left Shift: *Get First Set Bit Given an integer, find the position of the first set-bit (1) from the right.*

Input: n = 18

18 in binary = 0b10010

Output: 2

Procedure:

The logic of this solution relies on a combination of left shifting and the AND operation.

Essentially, we first check if the rightmost significant bit is the set bet using bit & 1. If not, we keep shifting left and checking until we find the bit that makes our AND operation yield 1.

The number of shifts is tracked by our pointer, k. Once we do find the set bit, we return k as our

answer.

Program

```
class FirstSetBitPosition
{
    private static int helper(int n)
    {
        if (n == 0)
        {
            return 0;
        }
        int p = 1;
        while (n > 0)
        {
            // Checking if last bit is set
            if ((n & 1) > 0) {
                return p;
            }

            // Increment position and right shift number
            p++;
            n = n >> 1;
        }
        return -1;
    }
    public static void main(String[] args)
    {
        System.out.println("First setbit position for number: 18 is -> " + helper(18));
        System.out.println("First setbit position for number: 5 is -> " + helper(5));
        System.out.println("First setbit position for number: 32 is -> " + helper(32));
    }
}
```

Output:

```
First setbit position for number: 18 is -> 2
First setbit position for number: 5 is -> 1
First setbit position for number: 32 is -> 6
```

Applications

- ✓ Counting Bits
- ✓ Palindrome Permutation
- ✓ Remove All Ones with Row and Column Flips
- ✓ Encode Number.
- ✓ Swapping Numbers
- ✓ Find the Non Repeated Elements in Array
- ✓ Minimum Bit Flips to Convert Number
- ✓ Longest Nice Subarray

1. Counting Bits

Given an integer n , return *an array ans of length $n + 1$ such that for each i ($0 \leq i \leq n$), $\text{ans}[i]$ is the **number of 1's in the binary representation of i .***

Input: $n = 2$

Output: [0,1,1] Explanation:

0 --> 0
1 --> 1
2 --> 10

Input: $n = 5$

Output: [0,1,1,2,1,2]

Explanation:

0 --> 0
1 --> 1
2 --> 10
3 --> 11
4 --> 100
5 --> 101

Procedure:

1. Since for decimal number 0 , no. of '1' bits in its binary representation = 0
2. For each value from 1 loop to iterate each bit of the number and count no. of '1' bits till the number becomes 0 i.e.it has no '1' bits remaining in its binary representation.
3. 0 will be added to count if last bit is 0 , and 1 will be added to count variable if last bit is 1,

thereby increasing count to 1.

4. Drop the last bit and repeat the process of counting the 1 bit.

Write a Java Program to count the number of 1 in bitrepresentation from 0 to a given value

CountingBits.java

```
import java.util.*;
class CountingBits
{
    public static int[] countBits(int n)
    {
        int r[]= new int[n+1];
        r[0]=0;
        for(int i=1; i<=n; i++)
        {
            int ct=0; int x=i;
            while(x>0)
            {
                x &= (x - 1);
                ct++;
            }
            r[i]= ct;
        }
        return r;
    }
    public static void main(String[] args)
    {
        Scanner s=new Scanner(System.in);
        int n=s.nextInt();
        int r[]=new int[n+1];
        r=countBits(n);
        System.out.print(Arrays.toString(r));}
}
```

2. Palindrome Permutation

Given a string, determine if a permutation of the string could form a palindrome.

Example 1:

Input: "code"

Output: false

Example 2:

Input: "aab"

Output: true

Example 3:

Input: "carerac"

Output: true

If a string with an even length is a palindrome, every character in the string must always occur even number of times. If the string with an odd length is a palindrome, every character except one of the characters must occur even number of times. Thus, in case of a palindrome, the number of characters with odd number of occurrences can't exceed 1 (1 in case of odd length and 0 in case of even length).

Procedure:

1. Retrieve characters from the string
 - a) Subtract ascii value of 'a' from the character (let's call it 'val').
 - b) Left shift on 1 'val' times
 - c) Perform XOR with a bitmask whose initial value is 0 after the left shift is performed
2. Repeat step 1 till the end of the string.
3. If the same character is seen next time bitmask value will be zero.
4. If there are even number of characters in the string and if it can be a palindrome the bitmask value will be zero
5. If there are odd number of characters and if it can be a palindrome the bitmask value with a bitwise & operation on bitmask-1 will be zero
6. If step 5 or step 6 is satisfied return true otherwise return false.

Write a Java Program to determine if a permutation of a string is a palindrome or not

PermutePalindrome.java

```
import java.util.*;
class PermutePalindrome
{
    static boolean canPermutePalindrome(String s)
    {
        int bitmask = 0;
        for(int i=0;i<s.length();i++)
        {
            char ch=s.charAt(i);
            int val = ch - 'a';
            bitmask = bitmask ^ (1 << val);
        }
        return (bitmask == 0 || (bitmask & (bitmask-1)) == 0 );
    }
    public static void main(String[] args)
```

```

{
    Scanner s=new Scanner(System.in);
    System.out.println("Enter the string");
    String ps=s.next();
    System.out.println(canPermutePalindrome(ps));
}
}

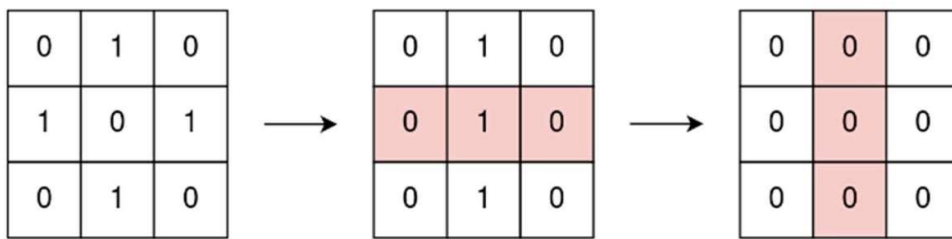
```

3. Remove All Ones with Row and Column Flips

We are given an $m \times n$ binary matrix grid.

In one operation, you can choose **any** row or column and flip each value in that row or column (i.e., changing all 0's to 1's, and all 1's to 0's).

Return true if it is possible to remove all 1's from the grid using any number of operations or false otherwise.



Input: grid = [[0,1,0], [1,0,1], [0,1,0]]

Output: true

Explanation: One possible way to remove all 1's from grid is to:

- Flip the middle row
- Flip the middle column

Procedure:

1. Compare first row with every row of the matrix.
2. If every row values are equal to first row or reversed row then it is possible to remove all 1's in the rows by row and column flips

Java Program to determine whether it is possible to remove all 1's from a binary matrix with row and column flips.

```
import java.util.*;
class RowColFlips
{
    public boolean removeOnes(int[][] grid)
    {
        // if the matrix only has one row then it is always true
        if (grid.length <= 1) return true;

        // start with the second row compare all values to the first row
        for (int i=1; i<grid.length; i++)
        {
            if (!equalsOrComplement(grid[0], grid[i]))
                return false;
        }
        return true;
    }

    private boolean equalsOrComplement(int[] row1, int[] otherrow)
    {
        boolean equals = true, complement = true;
        /*Every row should be equal to first row or complement to first row
        or else there will always be an extra value*/

        for (int i=0; i<row1.length; i++)
        {
            // If exclusive OR returns 1 it means both values are different
            complement = complement && ((first[i] ^ other[i]) == 1);

            // If exclusive OR returns 0 it means both values are same
            equals = equals && ((first[i] ^ other[i]) == 0);
        }
        return equals || complement;
    }

    public static void main(String args[])
    {
        Scanner sc=new Scanner(System.in);
        int m=sc.nextInt();
        int n=sc.nextInt();
        int grid[][]=new int[m][n];
        for(int i=0;i<m;i++)
            for(int j=0;j<n;j++)
                grid[i][j]=sc.nextInt();

        System.out.println(new RowColFlips().removeOnes(grid));
    }
}
```

```
}  
}
```

4. Encode Number

Given a non-negative integer **num**, Return its encoding string.

The encoding is done by converting the integer to a string using a secret function that you should deduce from the following table:

N	f(n)
0	""
1	"0"
2	"1"
3	"00"
4	"01"
5	"10"
6	"11"
7	"000"

Example 1:

Input: num = 23

Output: "1000"

Example 2:

Input: num = 107

Output: "101100"

If n is 0, then $f(n)$ is "". If $1 \leq n < 3$, then $f(n)$ is a binary string with length 1. If $3 \leq n < 7$, then $f(n)$ is a binary string with length 2. If $7 \leq n < 15$, then $f(n)$ is a binary string with length 3.

Procedure

1. Add one to the given number, then convert it to a binary string and remove the highest bit 1.

Write a java program to display the encoded form of a given string.

EncodeNumber.java

```
import java.util.*;
class EncodeNumber
{
    public String encode(int num)
    {
        String binaryString=Integer.toBinaryString(num+1);
        System.out.println("Binary String "+binaryString);
        return binaryString.substring(1);
    }

    static String encode2(int n)
    {
        n=n+1;
        StringBuilder sb=new StringBuilder("");
        while(n!=0)
        {
            sb.append(n&1);
            n=n>>1;
        }
        sb.reverse();
        System.out.println("Binary string " + sb.toString());
        return sb.toString().substring(1);
    }

    public static void main(String args[])
    {
        Scanner sc=new Scanner(System.in);
        int num=sc.nextInt();
        System.out.println(new EncodeNumber().encode(num));
        System.out.print(new EncodeNumber().encode2(num));
    }
}
```

5. Swapping numbers:

Given two variables a and y, swap two variables without using a third variable.

Examples:

Input: a = 2, b = 3

Output: a = 3, b = 2

Input: a = 20, b = 0

Output: a = 0, b = 20

Input: a = 10, b = 10

Output: a = 10, b = 10

Using Bitwise XOR

The idea is to use the properties of XOR to swap the two variables.

- **a = a ^ b:** Store the Bitwise XOR of **a** and **b** in **a**. Now, **a** holds the result of **(a ^ b)**.
- **b = a ^ b:** Bitwise XOR the new value of **a** with **b** to get the original value of **a**. This gives us, $b = (a \wedge b) \wedge b = a$.
- **a = a ^ b:** Bitwise XOR the new value of **a** with the new value of **b** (which is the original **a**) to get the original value of **b**. This gives us, $a = (a \wedge b) \wedge a = b$.

Time Complexity: O(1)

Auxiliary Space: O(1)

Write a java program to swap numbers without using a third variable

SwapUsingXOR.java

```
import java.util.Scanner;
class SwapUsingXOR
{
    static int a, b;
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        a = sc.nextInt();
        b = sc.nextInt();
        System.out.println("\nBefore Swapping:");
        System.out.println("a = " + a + ", b = " + b);
        swap();
        System.out.println("\nAfter Swapping:");
        System.out.println("a = " + a + ", b = " + b);
        sc.close();
    }

    static void swap()
    {
        // XOR Swapping
        a = a ^ b;
        b = a ^ b;
        a = a ^ b;
    }
}
```

6. Find the non-repeated Elements from the given an array**Case-1: Find the single non-repeated element**

Given an array of integers where: All elements appear exactly twice
One element appears only once (non-repeated)
Find the single non-repeated element using bit manipulation (XOR).

We discussed this earlier. Refer above notes.

Case-2: Find the two Non-Repeated Elements in the given Array

Given an array of integers where: All elements appear exactly twice
Two elements appear only once (non-repeated)
Find the two non-repeated elements using bit manipulation (XOR).

Example 1:

Input: nums = [2,2,1,3]

Output: [1,3]

Example 2:

Input: nums = [4,1,2,1,2,5]

Output: [4,5]

Algorithm:

- 1.Read array size and elements.
- 2.Compute XOR of all array elements.
- 3.Find rightmost set bit of the XOR result.
- 4.Divide array elements into two groups based on this bit. One group contains elements where this bit is set, another group contains elements where this bit is not set.
- 5.XOR elements in each group separately.
- 6.Output the two non-repeated elements.

Important XOR Properties:

$x \wedge x = 0 \rightarrow$ duplicates cancel each other

$x \wedge 0 = x$

XOR is commutative and associative

Step-by-Step Explanation

Example Input Array = [2, 4, 7, 9, 2, 4]

◆ Step 1: XOR all elements of the array

$\text{xor} = 2 \wedge 4 \wedge 7 \wedge 9 \wedge 2 \wedge 4$

Duplicate elements cancel:

$= (2 \wedge 2) \wedge (4 \wedge 4) \wedge 7 \wedge 9$

$= 0 \wedge 0 \wedge 7 \wedge 9$

$= 7 \wedge 9$

$= 13$

$\text{xor} = a \wedge b$ (where a and b are the two non-repeated elements)

◆ Step 2: Find the rightmost set bit of xor

$\text{rightMostSetBit} = \text{xor} \& (\sim\text{xor} + 1);$

☞ This bit is different in a and b

It helps us separate the two unique numbers.

◆ Step 3: Divide elements into two groups

Group 1: Numbers where this bit is set

Group 2: Numbers where this bit is not set

Since duplicate numbers are identical, they go into the same group and cancel out.

◆ Step 4: XOR elements in each group

XOR of Group 1 → gives first non-repeated number

XOR of Group 2 → gives second non-repeated number

Final Result:

Non-repeated elements are: 7 and 9

Write a java program to find the Two non-repeated elements:

```
import java.util.Scanner;
```

```
public class FindTwoNonRepeatedElements {
```

```
    // Method containing complete logic
```

```
    public static void findTwoNonRepeated(int[] arr) {
```

```
        // Step 1: XOR of all elements
```

```
        int xor = 0;
```

```
        for (int num : arr) {
```

```
            xor ^= num;
```

```
        }
```

```
        // Step 2: Find rightmost set bit
```



```
int rightMostSetBit = getRightMostSetBit(xor);
```

```
// Step 3: Divide elements into two groups
```

```
int num1 = 0, num2 = 0;
```

```
for (int num : arr) {
```

```
    if ((num & rightMostSetBit) == 0)
```

```
        num1 ^= num;
```

```
    else
```

```
        num2 ^= num;
```

```
}
```

```
// Step 4: Print result
```

```
System.out.println(num1 + " " + num2);
```

```
}
```

```
// First bit set
```

```
static int getRightMostSetBit(int n)
```

```
{
```

```
    // If no set bit
```

```
    if (n == 0) return 0;
```

```
    int res = 1;
```

```
    // Right shift 1 and check each bit
```

```
    while ((n & 1) == 0) {
```

```
        n = n >> 1;
```

```
        res++;
```

```
    }
```

```
    return res;
```

```
}
```

```
public static void main(String[] args) {
```

```
    Scanner sc = new Scanner(System.in);
```

```
    // Read array size
```

```
    int n = sc.nextInt();
```

```
    int[] arr = new int[n];
```

```
    // Read array elements
```

```
    for (int i = 0; i < n; i++) {
```

```
        arr[i] = sc.nextInt();
```

```
    }
```

```
    findTwoNonRepeated(arr);
```

```
    sc.close();
```

```
}  
}
```

7. Minimum Bit Flips to Convert Number

A bit flip of a number x is choosing a bit in the binary representation of x and flipping it from either 0 to 1 or 1 to 0.

For example, for $x = 7$, the binary representation is 111 and we may choose any bit (including any leading zeros not shown) and flip it.

We can flip the first bit from the right to get 110, flip the second bit from the right to get 101, flip the 3rd bit from the right (a leading zero) to get 011, etc.

Given two integers start and goal, return the minimum number of bit flips to convert start to goal.

Example 1:

Input: start = 10, goal = 7

Output: 3

Explanation: The binary representation of 10 and 7 are 1010 and 0111 respectively.

We can convert 10 to 7 in 3 steps:

- Flip the first bit from the right: 1010 -> 1011.
- Flip the third bit from the right: 1011 -> 1111.
- Flip the fourth bit from the right: 1111 -> 0111.

It can be shown we cannot convert 10 to 7 in less than 3 steps. Hence, we return 3.

Example 2:

Input: start = 3, goal = 4

Output: 3

Explanation: The binary representation of 3 and 4 are 011 and 100 respectively. We can convert 3 to 4 in 3 steps:

- Flip the first bit from the right: 011 -> 010.
- Flip the second bit from the right: 010 -> 000.
- Flip the third bit from the right: 000 -> 100.

It can be shown we cannot convert 3 to 4 in less than 3 steps. Hence, we return 3.

Intuition

-> The question asks how few bit flips are necessary to change one integer into another.

-> It follows easily that the bits of the two numbers can be compared.

The amount of flips required will depend on how the bits in the binary representation of start and aim differ from one another.

->The result of a bitwise XOR operation between start and goal is a number where each 1 denotes a location where the bits of the two numbers differ.

->The necessary number of bit flips can be found by counting the 1s in this XOR result.

Approach

1.XOR the two numbers: Perform a bitwise XOR between start and goal.

The result will have 1 in positions where the bits are different between the two numbers.

2.Count the number of 1s in the XOR result: This tells us the number of positions where the bits differ and hence the number of bit flips required.

3.Bit counting: Use a loop to repeatedly check the last bit of the XOR result using & 1.

Right shift the XOR result ($>> 1$) to process the next bit. Continue until all bits are processed.

Time complexity:

The time complexity is $O(k)$, where k is the number of bits in the binary representation of the larger of the two numbers (start or goal).

Space complexity:

The space complexity is $O(1)$.

Step by step Tracing:

Example: Input start = 10 goal = 7

Step 1: Convert to Binary

```
-----  
Number  Decimal  Binary  
start   10      1010  
goal    7        0111
```

(We align bits from the right)

Step 2: XOR Operation

```
-----  
int xor = start ^ goal;
```

Binary XOR

```
  1010 (10)  
^ 0111 (7)
```

```
-----  
  1101
```

Decimal: xor = 13

◆ Meaning: Each 1 in 1101 represents a position where bits differ → one flip needed.

Step 3: Initialize Variables

```
-----  
ans = 0  
xor = 13 (1101)
```

Step 4: While Loop Tracing

```
-----  
↻ Iteration 1  
xor = 1101
```

```

xor & 1 = 1
ans = 0 + 1 = 1
xor >>= 1 → 0110

```

↻ Iteration 2

```

xor = 0110
xor & 1 = 0
ans = 1
xor >>= 1 → 0011

```

↻ Iteration 3

```

xor = 0011
xor & 1 = 1
ans = 1 + 1 = 2
xor >>= 1 → 0001

```

↻ Iteration 4

```

xor = 0001
xor & 1 = 1
ans = 2 + 1 = 3
xor >>= 1 → 0000

```

Loop Ends (xor == 0)

✓ Final Output

```

ans = 3

```

✓ Minimum bit flips required = 3

Visual Summary Table

Iteration	xor (binary)	xor & 1	ans	xor >> 1
1	1101	1	1	0110
2	0110	0	1	0011
3	0011	1	2	0001
4	0001	1	3	0000

Write a java program to find the Minimum Bit Flips to Convert Number

MinBitFlipstoConvertNumber.java

```

import java.util.Scanner;

```

```

class MinBitFlipstoConvertNumber
{
    public int minBitFlips(int start, int goal)
    {

```

```
int ans = 0;
// XOR will give 1 where the bits are different
int xor = start ^ goal;

while(xor!=0)
{
    // If the last bit is 1, increment count
    ans += xor & 1;

    // Right-shift to check the next bit
    xor >>=1;
}
return ans;
}
public static void main(String[] args)
{
    Scanner sc=new Scanner(System.in);
    int start=sc.nextInt();
    int goal=sc.nextInt();
    System.out.println(new MinBitFlipstoConvertNumber().minBitFlips(start,goal));
}
}
```

8. Longest Nice Subarray

You are given an array nums consisting of positive integers.

We call a subarray of nums nice if the bitwise AND of every pair of elements that are in different positions in the subarray is equal to 0.

Return the length of the longest nice subarray.

A subarray is a contiguous part of an array.

Note that subarrays of length 1 are always considered nice.

Example 1:

Input: nums = [1,3,8,48,10]

Output: 3

Explanation: The longest nice subarray is [3,8,48]. This subarray satisfies the conditions:

- 3 AND 8 = 0.

- 3 AND 48 = 0.

- 8 AND 48 = 0.

It can be proven that no longer nice subarray can be obtained, so we return 3.

Example 2:

Input: nums = [3,1,5,11,13]

Output: 1

Explanation: The length of the longest nice subarray is 1. Any subarray of length 1 can be chosen.

Constraints: $1 \leq \text{nums.length} \leq 105$ $1 \leq \text{nums}[i] \leq 109$ **Intuition:**

We need to find the longest "nice" subarray, where:

The bitwise AND of any two elements in the subarray should be zero (i.e., no overlapping bits).

This means no two numbers in the subarray should share a '1' bit in the same position.

To solve this, we use a sliding window approach:

Expand the window (r) while ensuring the subarray remains "nice".

If the condition is violated, shrink the window (l) until the subarray is valid again.

Approach:**1.Sliding Window Technique:**

Use two pointers (l and r) to define a valid subarray.

Maintain a bitmask (usedBits) to track the OR result of numbers in the current window.

If adding $\text{nums}[r]$ causes a conflict ($\text{usedBits} \& \text{nums}[r] \neq 0$), shrink the window from the left (l).

Otherwise, expand the window and update maxLength.

2.Bitwise Operations:

$\text{usedBits} \mid= \text{nums}[r] \rightarrow$ Adds the new number's bits to the mask.

$\text{usedBits} \wedge= \text{nums}[l] \rightarrow$ Removes $\text{nums}[l]$ when shrinking the window.

Complexity Analysis

Time Complexity: $O(N)$

Each element is added once and removed once $\rightarrow O(2N) = O(N)$.

Space Complexity: $O(1)$

We only use integer variables.

STEP-BY-STEP TRACE

l (left pointer)

r (right pointer)

mask (in binary)

Conflict check

Window size

Example Input

$\text{nums} = [1, 3, 8, 48, 10]$

Binary Representation:

Number Binary

1 000001

3 000011

8 001000

48 110000

10 001010

Initial State

mask = 000000

ans = 0

l = 0

Iteration-by-Iteration Trace

◆ r = 0 → nums[0] = 1 (000001)

Conflict Check: & → detects overlapping bits

mask & nums[r]

000000 & 000001 = 000000 → NO conflict

Add to mask: mask always stores OR of current window

mask |= nums[r]

000000 | 000001 = 000001

Update ans

window size = r - l + 1 = 0 - 0 + 1 = 1

ans = 1

Window: [1]

◆ r = 1 → nums[1] = 3 (000011)

Conflict Check:

000001 & 000011 = 000001 ≠ 0 → CONFLICT

✗ Conflict found → shrink window //^ → removes left element bits safely

Remove nums[l] = 1

mask ^= nums[l]

000001 ^ 000001 = 000000

l = 1

Check again

$000000 \& 000011 = 000000 \rightarrow \text{NO conflict}$

Add nums[r]

mask |= 000011

$000000 \mid 000011 = 000011$

Update ans

window size = $1 - 1 + 1 = 1$

ans = 1

✓ Window: [3]

◆ $r = 2 \rightarrow \text{nums}[2] = 8 \text{ (001000)}$

Conflict Check:

$000011 \& 001000 = 000000 \rightarrow \text{NO conflict}$

Add to mask

$000011 \mid 001000 = 001011$

Update ans

window size = $2 - 1 + 1 = 2$

ans = 2

✓ Window: [3, 8]

◆ $r = 3 \rightarrow \text{nums}[3] = 48 \text{ (110000)}$

Conflict Check:

$001011 \& 110000 = 000000 \rightarrow \text{NO conflict}$

Add to mask:

$001011 \mid 110000 = 111011$

Update ans

window size = $3 - 1 + 1 = 3$

ans = 3

✓ Window: [3, 8, 48]

◆ $r = 4 \rightarrow \text{nums}[4] = 10 \text{ (001010)}$

Conflict Check:

$111011 \& 001010 = 001010 \neq 0 \rightarrow \text{CONFLICT}$

✗ Conflict $\rightarrow$ shrink window

Remove $\text{nums}[l] = 3$

$\text{mask} \wedge = 000011$

$111011 \wedge 000011 = 111000$

$l = 2$

Check again

$111000 \& 001010 = 001000 \neq 0 \rightarrow \text{CONFLICT}$

Remove $\text{nums}[l] = 8$

$\text{mask} \wedge = 001000$

$111000 \wedge 001000 = 110000$

$l = 3$

Check again

$110000 \& 001010 = 000000 \rightarrow \text{NO conflict}$

Add $\text{nums}[r]$

$110000 \mid 001010 = 111010$

Update ans

window size = $4 - 3 + 1 = 2$

ans = 3

✓ Window: [48, 10]

✓ Final Answer

ans = 3

Longest Nice Subarray = [3, 8, 48]

Write a java program to find the length of the longest nice subarray

LongestNiceSubarray.java

```
import java.util.Scanner;
```

```
class LongestNiceSubarray
```

```
{
```

```
    public int longestNiceSubarray(int[] nums)
```

```
    {
```

```
        int ans = 0;
```

```
        int mask = 0;
```

```
        for (int l = 0, r = 0; r < nums.length; r++)
```

```
        {
```

```
            // If conflict exists, shrink window
```

```
        while ((mask & nums[r]) != 0)
        {
            mask ^= nums[l];
            l++;
        }

        // Add current number bits
        mask |= nums[r];

        // Update maximum length
        ans = Math.max(ans, r - l + 1);
    }
    return ans;
}

public static void main(String[] args)
{
    Scanner sc=new Scanner(System.in);
    int n=sc.nextInt();
    int a[]=new int[n];

    for(int i=0 ;i<n;i++)
    {
        a[i]=sc.nextInt();
    }
    System.out.println(new LongestNiceSubarray().longestNiceSubarray(a));
}
}
```